RENEWABLE ENERGY GENERATION ANALYSIS

1. Project Overview and Objective

This project focuses of renewable energy performance using a comprehensive dataset of daily power generation across various Indian states and regions. By examining key attributes such as power station sector (Central, State, and Private) and station type (Solar, Wind, Hybrid) the study provides a broad view of the country's renewable energy production landscape.

The analysis compares installed (monitored) capacity with actual power generation to assess how effectively renewable power stations utilize their capacity. It also considers regional and seasonal variations in generation to understand differences in performance across locations. This project supports evaluation of operational efficiency and offers insights into India's progress toward sustainable energy and reliable power supply.

The primary objective of this project is to evaluate renewable energy generation patterns across various states and regions by analysing the relationship between installed capacity and actual power generation, and to identify performance differences among renewable power stations.

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2. Data Sources

- **Source Description and Timeline:** India Data Portal / Source -Ministry of Power/ Energy - 2019-2023
- **Domain**: Energy / Renewable Energy (India)
- **Link:** https://indiadataportal.com/p/power/r/mop-renewable_energy_generation-pl-dl-aaa

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3. Problem Statement

- To analyse state-wise distribution of installed renewable energy capacity and actual power generation.
- To compare renewable energy generation across different regions (North, South,West).
- To evaluate the contribution of different renewable energy sources (Solar, Wind, Hybrid) based on installed capacity and actual generation.
- To analyse the performance of power stations and solar parks by comparing installed capacity with actual generation.
- To study sector-wise contribution (Public vs Private) in renewable energy generation.
- To study the trend of renewable energy generation over time using date-wise data.
- To analyse the gap between installed capacity and actual generation to understand utilization efficiency.
- Seasonal patterns will be identified by deriving seasons from the date field to understand variations in energy generation.

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4. Attribute (Column /Features) Details:

Fact Table Details:

Column Name	Data Type	Description
Energy_Id	Text(String-Number)	Unique identifier for each energy generation record.
Date	Date	Date on which the energy was generated.
Season	Text (String)	Indicates the season (Summer, Monsoon, Winter) during which the data was recorded.
State_Code	Text (String-Number)	Code representing the state where the station is located (example: GU_24, KA_29).
Station_Id	Text (String-Number)	Unique ID of the power generation station (linked with Station Dimension table).
Installed_Capacity	Decimal Number	Maximum power generation capacity of the station (in MW).
Actual_Generation	Decimal Number	Actual amount of energy generated by the station.
Efficiency	Decimal Number	Percentage of installed capacity that was actually utilized for power generation.
Unused Capacity	Text (String)	Difference between installed capacity and actual generation, showing unused potential.
Efficiency_Category	Text (String)	Performance classification based on efficiency (Very Low, Low, Medium, High)

Location_Dim Details:

Column Name	Data Type	Description
State_Code	Text (String-Number)	Official code representing each state for identification.
State_Name	Text (String)	Name of the state where the power station is located.
Region	Text (String)	Geographical region of the state (North, South, East)
Solar_Park	Text (String)	Name of the solar park where the power station is situated.

Station_Dim Details:

Column Name	Data Type	Description
Station_Id	Text (String-Number)	Unique identifier assigned to each power generation station.
Station_Name	Text (String)	Name of the power generation company or station (example: Giwel Energy, Ostro Energy).
Sectore	Text (String)	Type of sector to which the station belongs (Private, Central, or State).
Owner	Text (String)	Ownership category of the station such as IPP (Independent Power Producer) or Government
Energy_Type	Text(String)	Type of renewable energy generated by the station (Solar, Wind, Hybrid)

5. Tools & Technologies

- **Excel:** Data cleaning, Transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing in Excel

Tasks Performed:

Data Cleaning & Transformation: Fact Table

The original dataset consisted of **200,235 records**; however, due to **memory efficiency constraints**, a subset of **100,903 records** was processed for data cleaning.

Energy_Id Standardised:

- Identified records where the **ID column contained values starting from 0 and appearing as standalone entries**.
- Imputed the corresponding values in the **EnerId_1** column to handle inconsistent identifiers.
- Applied **TRIM ()** function to remove leading and trailing spaces.
- Applied **CLEAN ()** function to remove non-printable and unwanted characters.
- Ensured the column data type was set to **Text** to maintain consistency and prevent unintended type conversion.

Date Standardised:

- Converted the Date column to **Date data type**.
- Applied **TRIM ()** function to remove leading and trailing spaces.
- Applied **CLEAN ()** function to remove non-printable and unwanted characters.

Season Standardised:

- The Season column is created to categorize each record based on the month of the date.
- It helps in analysing data season-wise instead of only date-wise.
- This improves understanding of seasonal trends and patterns.

Formula

```
=IF(OR(MONTH(DATEVALUE(B2))=2,MONTH(DATEVALUE(B2))=3),"Spring",IF(OR(MONTH(DATEVALUE(B2))=4,MONTH(DATEVALUE(B2))=5,MONTH(DATEVALUE(B2))=6),"Summer",IF(OR(MONTH(DATEVALUE(B2))=7,MONTH(DATEVALUE(B2))=8,MONTH(DATEVALUE(B2))=9),"Monsoon","Winter")))
```

State_Code Standardised:

- The **State_Code** is used as a **foreign key** in the **Location** dimension table to establish a relationship with the corresponding state records in the fact table.
- This key ensures referential integrity and enables accurate mapping between transactional data and state-level location attributes such as **State_Name, Region, and Solar_Park**.

Formula =VLOOKUP(C2,Location!$A$1:$B$9,1,FALSE)

- Applied the **TRIM()** and **CLEAN()** functions to remove extra spaces and non-printable characters, and ensured that the data type was set to **Text** for consistency and accurate processing.

Station_Id Standatdised:

- The Station_Id is used as a unique identifier to distinguish each power generation station in the fact table.
- Applied the **TRIM()** and **CLEAN()** functions to remove extra spaces and non-printable characters from the Station_Id values.
- The data type was set to **Text** to ensure consistency and prevent mismatches during joins and filtering.

Installed Capacity Standardised:

- Missing values were imputed using the average Installed_Capacity of the same category to preserve data consistency and reduce bias in analysis**.**
- This field is used for evaluating potential output and comparing station capacity across locations.
- Applied the TRIM() and CLEAN() functions where required to eliminate hidden characters from numeric fields imported as text.
- The data type was converted to Decimal Number to allow accurate calculations and aggregations.

Formula=IF(ISBLANK(K3647),AVERAGEIF($J$2:$J$3646,J3647, $K$2:$K$3646),K3647)

Actual_Generation Standardised:

- Negative values identified in the Actual_Generation column were considered data anomalies, as power generation cannot be physically negative.
- These values were corrected by converting them to positive and the transformation was documented to maintain analytical integrity.
- Zero generation values, which may indicate plant shutdowns or missing measurements, were treated as missing data. These values were imputed using the **average generation of the corresponding Energy_Type (e.g., wind or solar)** to preserve continuity in the dataset and to avoid skewing the analysis with non-operational periods.
- Missing values were imputed using the average Actual_Generation Capacity of the same category to preserve data consistency and reduce bias in analysis**.**

Formula =IF(M2=0,AVERAGEIF($J$2:$J$3646, J2, $M$2:$M$3646),M2)

- Applied the TRIM() and CLEAN() functions to remove unwanted spaces and non-printable characters from imported values.
- The data type was set to Decimal Number to ensure precision in analytical calculations.

Efficiency Standardised:

- The **Efficiency** column was derived to measure the utilization of installed capacity.
- Efficiency was calculated using the
- Data Cleaning was performed using TRIM() and CLEAN() to ensure reliable input values before calculation.
- The data type was set to Decimal Number to ensure precision in analytical calculations.

```
Formula= M2/K2*100
Actual_Generation/ Installed_Capacity *100
```

Unused Capacity:

- The Unused Capacity represents the portion of installed capacity that was not utilized.
- Unused Capacity was calculated using the formula:Unused Capacity = Installed_Capacity − Actual_Generation.

```
Formula: =F2-G2
```

- Input fields were cleaned using TRIM() and CLEAN() to avoid calculation errors caused by hidden characters.
- The data type for the numeric columns was ensured as Number to support accurate calculations.

Efiiciency_Category:

- The **Efficiency_Category** was derived by classifying stations into **Very Low, Low, Medium, and High** categories based on the values in the **Efficiency** column to enable clear and standardized interpretation of operational efficiency.

```
Formula=IF(AND(H2>=0,H2<=1),"Very
Low",IF(AND(H2>=1,H2<=20),"Low",IF(AND(H2>20,H2<=40),"Medium",IF(AND(H2>40,H2<8
5),"High",""))))
```

- Input fields were cleaned using TRIM() and CLEAN() to avoid calculation errors caused by hidden characters.
- The data type was set to **Text** to ensure uniform formatting and reliable processing of the column values.

Data Cleaning :Location_Dim Table

State_Code Standardised:

- The State_Code column was cleaned using the TRIM() and CLEAN() functions to remove extra spaces and non-printable characters.
- The data type of the column was explicitly ensured as Text to maintain consistency and support accurate matching and processing.

State_Name Standardised:

- The **State_Name** column was standardized by applying the **TRIM()** and **CLEAN()** functions to remove extra spaces and non-printable characters.
- The data type of the column was explicitly ensured as **Text** to maintain consistency and support accurate grouping, filtering, and mapping operations.

```
Formula=VLOOKUP(A2,'Fact'!C2:M100904,2,FALSE)
```

Region Standardised:

- The **Region** column was standardized by applying the **TRIM()** and **CLEAN()** functions to remove extra spaces and non-printable characters.
- The data type of the column was explicitly ensured as **Text** to maintain consistency and support accurate classification, grouping, and regional analysis.

```
Formula=VLOOKUP(A2,'Fact'!$C$2:$F$100904,3,FALSE)
```

Solar_Park Standardised:

- The Solar_Park column was standardized by applying the TRIM() and CLEAN() functions to remove extra spaces and non-printable characters.
- The data type of the column was explicitly ensured as Text to maintain consistency and support accurate grouping, filtering, and mapping of solar park locations.

```
Formula=VLOOKUP(A2,'Fact’!C2:M100904,4,FALSE)
```

Data Cleaning :Station_Dim Table

Station_Id Standardised:

- Based on the cleaned and deduplicated station names, corresponding Station_ID values were imputed.
- The Station_Id column was standardized by applying the TRIM() and CLEAN() functions to remove extra spaces and non-printable characters.
- The data type of the column was explicitly ensured as Text to maintain consistency and enable accurate station-level identification and analysis.

Station_Name Standardised:

- The **Station_Name** column was standardized by applying the **TRIM()** and **CLEAN()** functions to remove extra spaces and non-printable characters.
- The data type of the column was explicitly ensured as **Text** to maintain consistency and support accurate station identification and reporting

Sector Standardised:

- The Sector column was standardized by applying the TRIM() and CLEAN() functions to remove extra spaces and non-printable characters.
- The data type of the column was explicitly ensured as Text to maintain consistency and support accurate categorization and analysis.

```
Formula=VLOOKUP(A2,Table1[[Station_Id]:[Type]],2,FALSE)
```

Owner Standardised:

- The **Owner** column was standardized by applying the **TRIM()** and **CLEAN()** functions to remove extra spaces and non-printable characters.
- The data type of the column was explicitly ensured as **Text** to maintain consistency and enable accurate grouping and reporting by owner.

```
Formula=VLOOKUP(A2,Table1[[Station_Id]:[Actual_Generation]],3,FALSE)
```

Energy_Type Standardised:

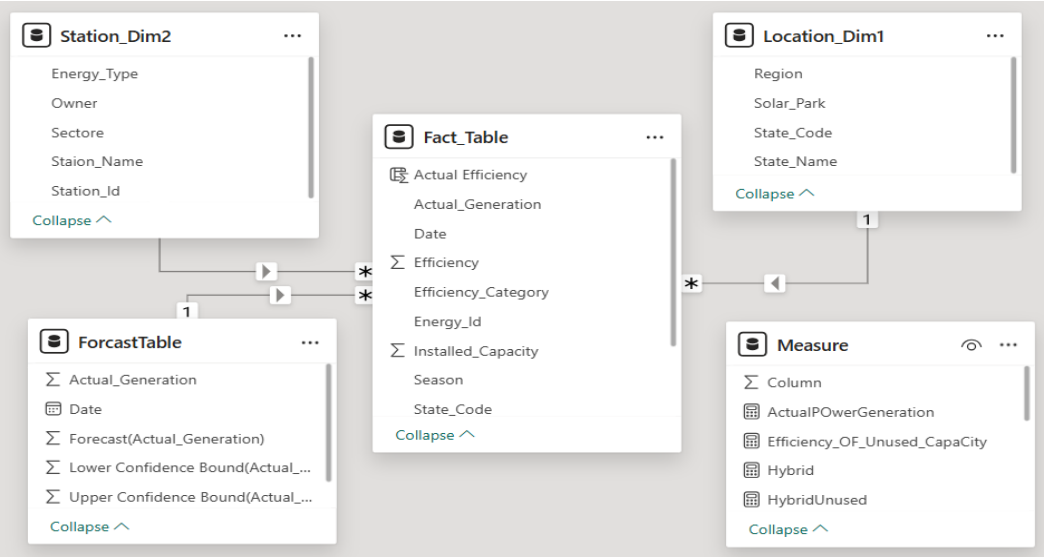
- The Energy_type column was standardized by applying the TRIM() and CLEAN() functions to remove extra spaces and non-printable characters.
- The data type of the column was explicitly ensured as Text to maintain consistency and enable accurate categorization and analysis by energy type.

```
Formula= =VLOOKUP(A2,Table1[[Station_Id]:[Type]],4,FALSE)
```

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

Data Model Insights



• The data model is designed using a **star schema** architecture, with a central **Fact_Table** connected to supporting **dimension tables** (Station_Dim and Location_Dim). This structure enables efficient analytical querying and simplified reporting.

• The **Fact_Table** contains key quantitative measures, including *Installed Capacity, Actual Generation, Efficiency,* and *Unused Capacity*. These metrics form the basis for performance and utilization analysis.

• The **Station_Dim** table stores descriptive attributes such as *Station Name, Sector, Owner,* and *Energy Type*, which facilitate detailed analysis based on station-level characteristics.

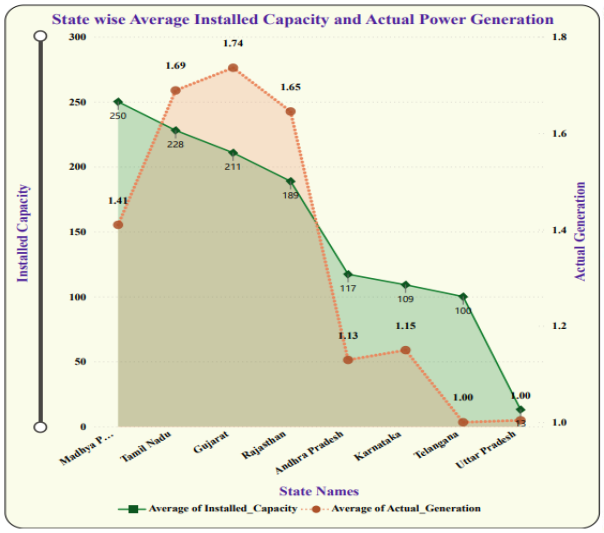
• The **Location_Dim** table captures geographical attributes including *State Code, State Name, Region,* and *Solar Park*, allowing for regional and location-based analysis.

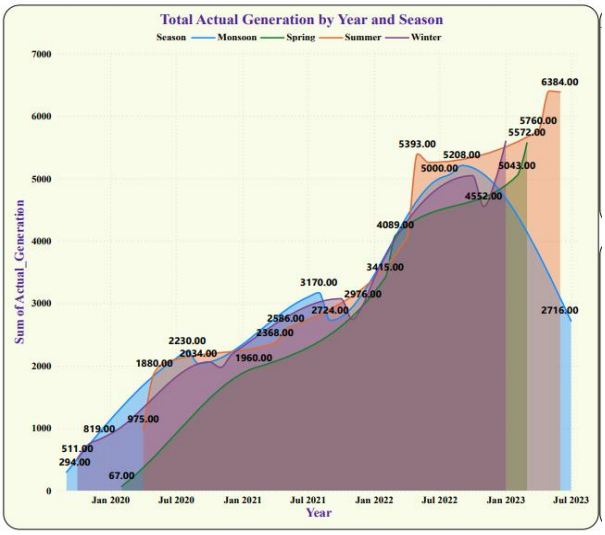
• A **one-to-many (1:*) relationship** exists between Station_Dim and Fact_Table, ensuring that each station can be associated with multiple fact records while maintaining data normalization.

• A **one-to-many (1:*) relationship** also exists between Location_Dim and Fact_Table, enabling multiple fact records to be linked to a single location for aggregation and trend analysis.

- **Forecasting metrics**: Actual_Generation, Forecast values, Upper/Lower Confidence Bounds
- Enables trend analysis and future projections (2020-2024)

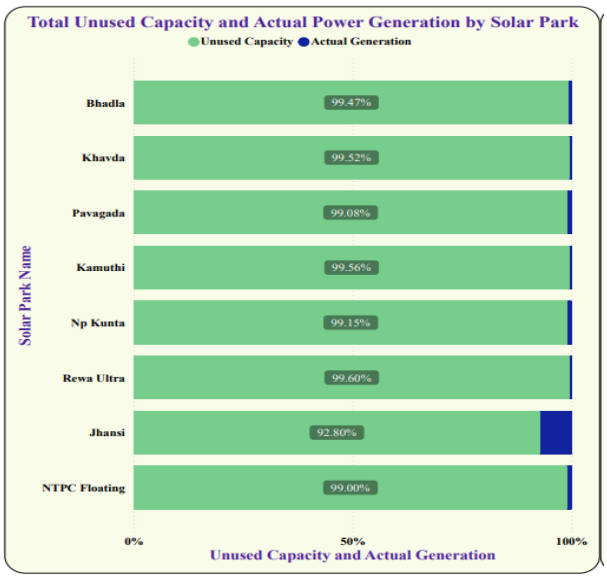
Insights From data Visualizations

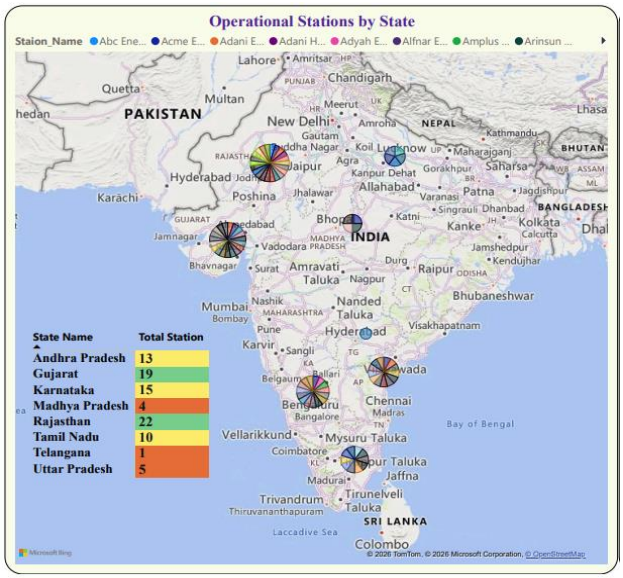




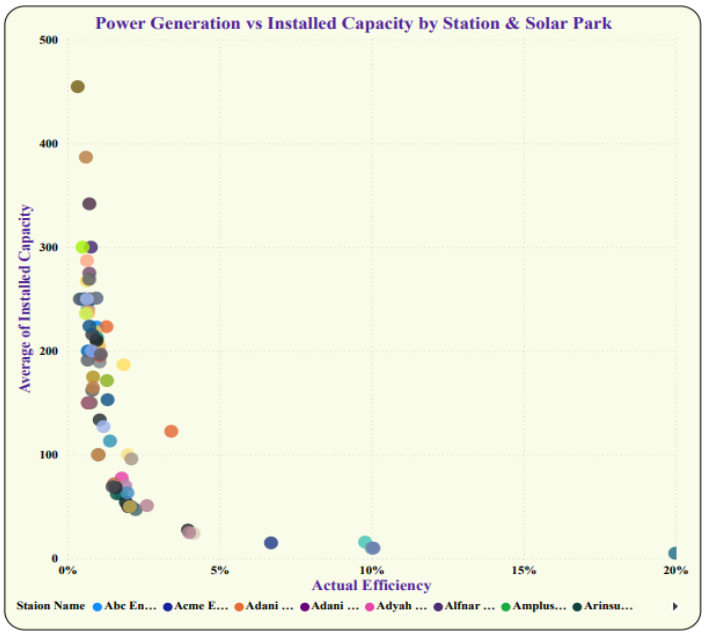
- Total Power Generation: 145,587 MW and Installed Capacity: 16,281,171 MW, showing very low utilization (average efficiency 1.89%).
- Unused Capacity: 16,162,419 MW, indicating large under-utilized potential across stations.
- Energy Mix: Solar dominates with 102295 MW, followed by Wind 43196 and Hybrid 96 MW.
- Geographical Spread: Rajasthan leads with 22 stations, followed by Gujarat (19) and Karnataka (15).
- Regional Generation: North region highest (59,523 MW), then South (52,468 MW) and West (33,596 MW).
- Seasonal Trend: Peak generation in Winter (45,759 MW) and Summer (43,803 MW); lowest during Monsoon (22,361 MW).
- Year_over_year growth: 2020(294 MW)→2023 (6384 MW)

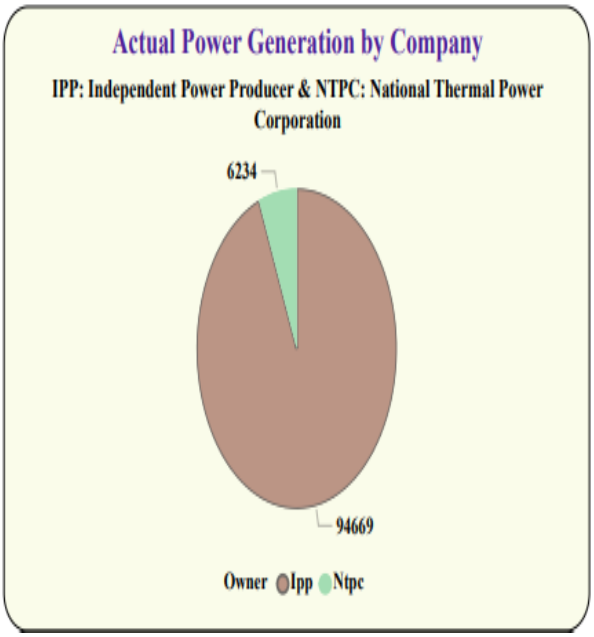
- Top Generators: Adani Energy (18,314 MW), Renew Energy (14,245 MW), and Acme Energy (9,823 MW).
- Solar Park Efficiency: Best performers are Rewa Ultra (99.60%) and Kamuthi (99.56%); lowest is Jhansi (92.80%).



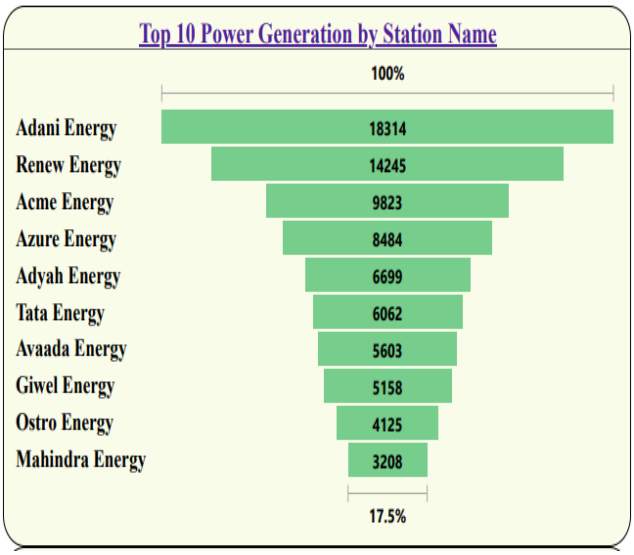


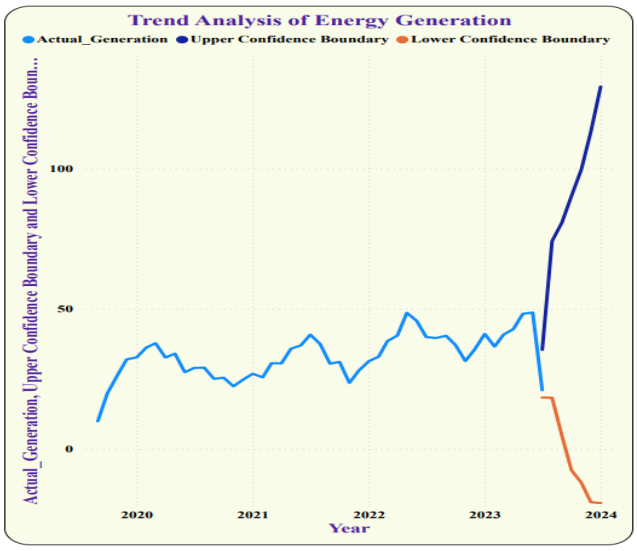
- ~99% unused capacity indicates significant infrastructure underutilization
- Solar parks show 92-99% unused capacity → potential grid connectivity or demand-side constraints
- Very Low efficiency plants contribute 145,158 MW, while High efficiency contributes only 69MW, revealing operational and technology gaps
- North region leads generation due to large solar parks (Bhadla, Khavda)
- South shows balanced solar-wind mix, improving reliability
- Monsoon dip: Reduced solar irradiance + maintenance activities
- Winter peak: Optimal solar conditions + higher demand
- IPPs generate 94,669 MW vs NTPC's 6,234 MW → private sector leading renewable adoption
- Top stations correlate with states having favourable policies (Gujarat, Rajasthan)
- Bottom 10 stations (e.g., Dadri, Singrauli) show <700 MW generation → aging infrastructure or resource constraints





- Persistent grid constraints will force generation curtailment during peak production hours, wasting available solar output.
- Solar parks will continue to miss ROI targets if grid bottlenecks, curtailment, and technical faults are not resolved, resulting in prolonged payback periods and wasted generation
- Lack of real-time forecasting will keep operations reactive, meaning failures will be addressed only after losses occur instead of being prevented
- Reactive maintenance increases unplanned outages, raising repair costs and reducing plant availability
- Financial risk will grow as revenue remains tied to unpredictable seasonal output rather than stable baseload generation
- Investor confidence may weaken due to repeated underperformance against projected capacity utilization factors (CUF)
- Delayed issue detection will amplify small technical faults into major shutdown events
- Operational inefficiencies will continue unless performance data is used for early-warning alerts and predictive maintenance
- Grid stress events will become more frequent as generation and evacuation remain misaligned
- Without analytics, operations will stay reactive, causing more unplanned outages, higher maintenance costs, and greater revenue loss





• Capture excess monsoon generation through battery storage systems so surplus energy can be supplied during extreme summer shortfalls

• Investigate the sharp generation decline between 2023 and 2024 by reviewing fuel supply records, maintenance logs, and weather conditions to identify root causes

• Reduce dependency on dominant producers such as Adani, ReNew, and ACME by promoting expansion of mid-tier firms like Azure Power and Adyah

• Conduct a technical and grid audit of all stations in Rajasthan to assess transmission readiness and resource utilization

• Perform performance audits on bottom-ranked stations such as Dadri Thermal Power Station and Singrauli Super Thermal Power Station and decide on repair, upgrade, or decommissioning within a defined timeline

• Identify plants classified as “High Efficiency” and prioritize capacity expansion so their contribution becomes grid-significant

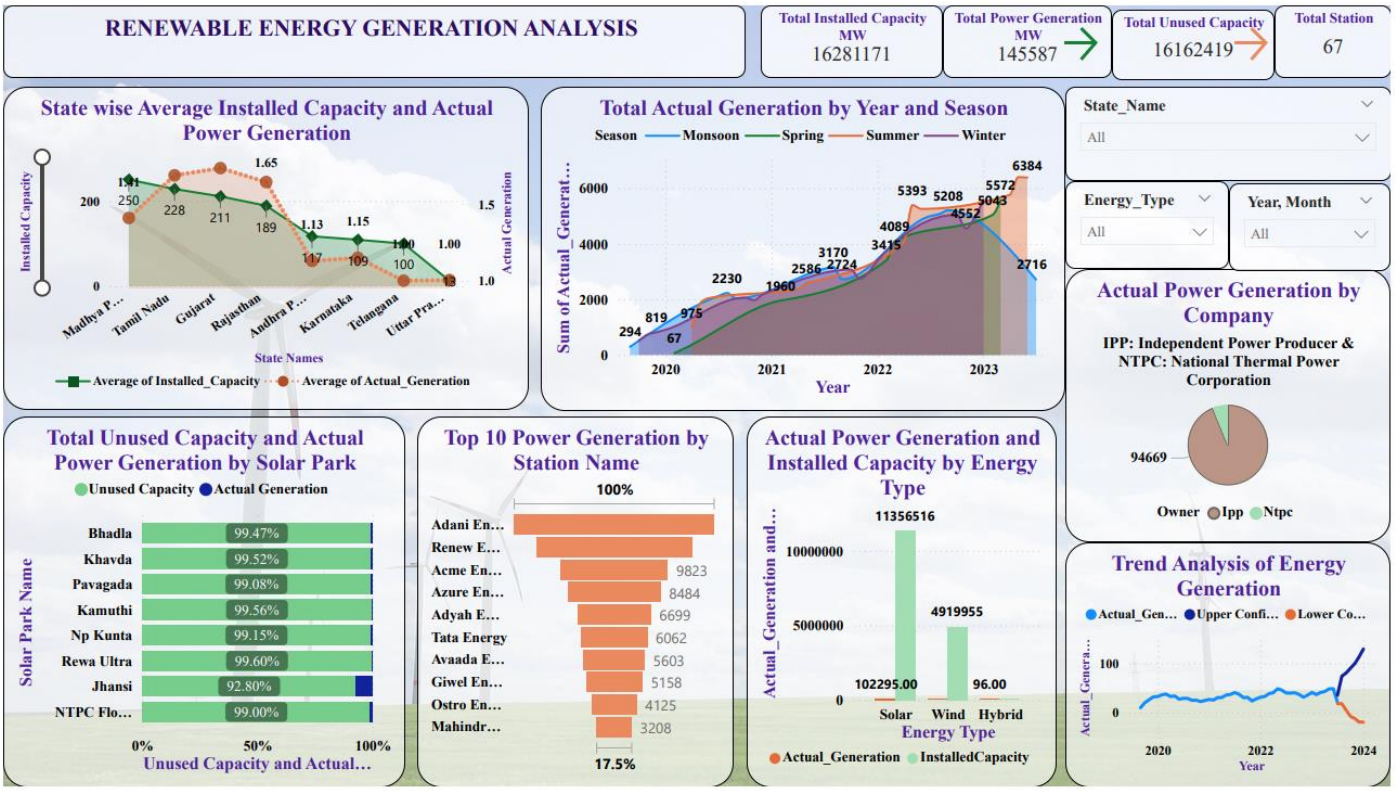
• Carry out on-site inspections of solar parks showing high unused capacity to detect grid connectivity issues, inverter faults, and panel degradation

• Pause new capacity installations until existing assets are better utilized and transmission infrastructure is strengthened

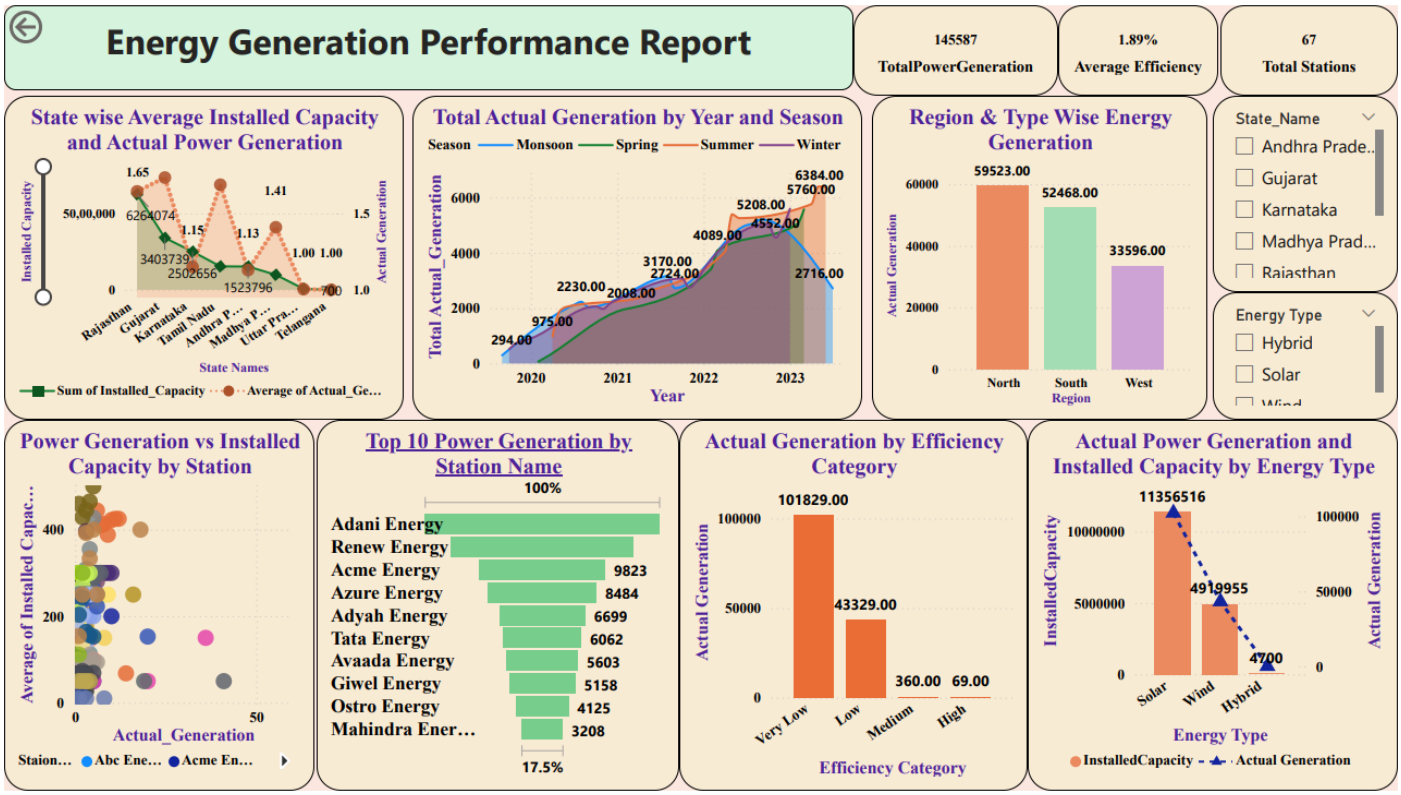
• Align maintenance schedules with low-output seasons and secure power purchase agreements to stabilize generation during weaker periods

• Integrate a predictive forecasting model using historical data to estimate future seasonal generation and support proactive planning

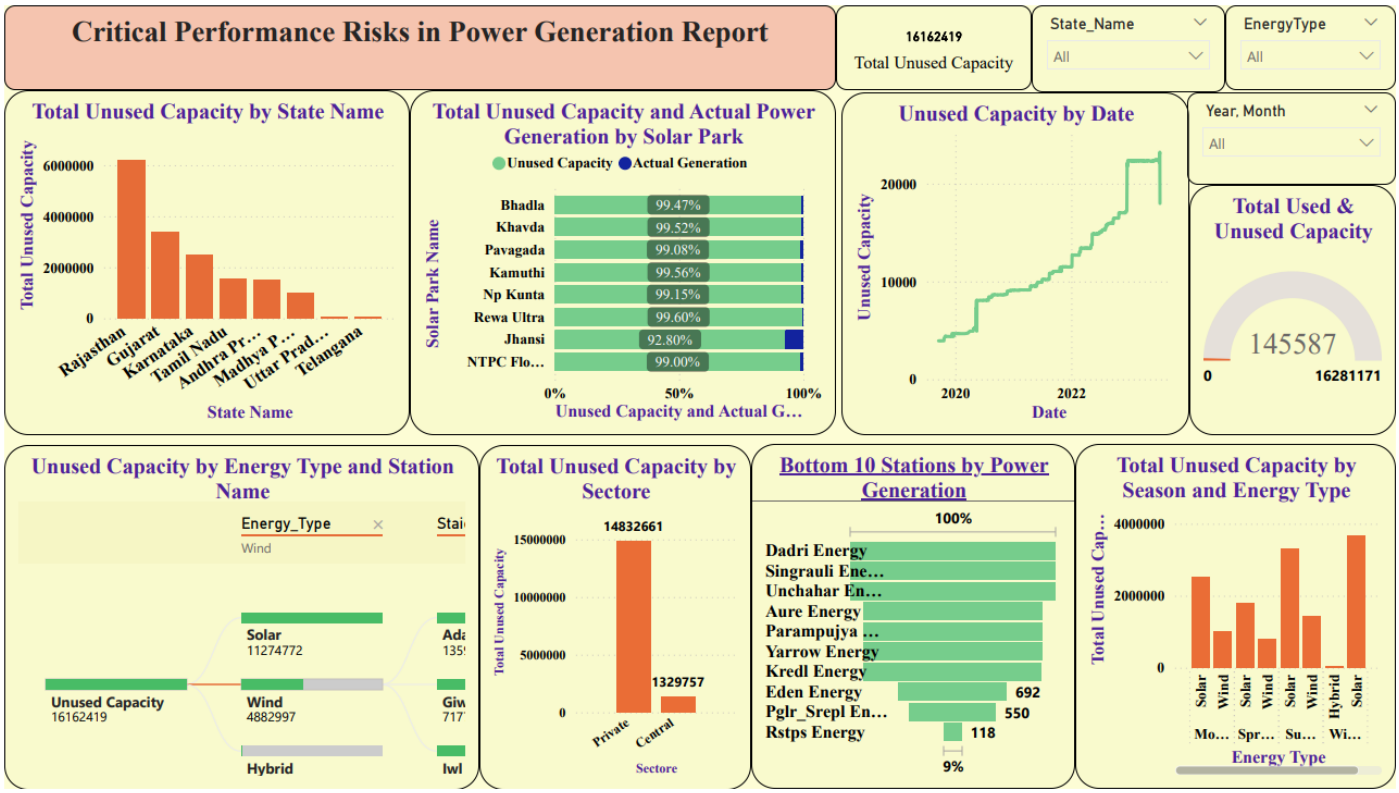
Dashboard



Energy Generation Performance Report



Critical Performance Risks in Power Generation Report



Conclusions

- Only 0.89% of installed capacity is producing power, indicating extreme underutilization
- Generation is sharply declining in 2023–2024, signaling a risk of collapse if trends continue
- Rajasthan has the most stations but weak generation output, showing poor regional performance
- Seasonal imbalance is critical: Monsoon output is extremely high, while Summer output is near zero
- Market is highly concentrated, with a few companies dominating generation
- Bottom-performing stations are close to non-viable, threatening asset value and grid stability